PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Outcomes of Hyperglycaemia in Pregnancy in Africa: Systematic Review Study Protocol
AUTHORS	Musa, Ezekiel; Chivese, Tawanda; Werfalli, Mahmoud; Matjila, Mushi; Norris, Shane; Levitt, Naomi

VERSION 1 – REVIEW

REVIEWER	Han Ting Wang Maisonneuve-Rosemont Hospital Montreal, Quebec, Canada
REVIEW RETURNED	03-Jun-2020

GENERAL COMMENTS	Dear Editorial board and authors. thank you for the opportunity to review this protocol. This is a protocol for a systematic review looking at the outcome of hyperglycemia in pregnancy. The subject in itself is indeed very important for a continent where access to healthcare might be quite different from most western countries. Overall the protocol is well written and all major aspect of the systematic review is present. therefore i only have some minor comments for the authors: 1. Outcomes. There are a lot of outcomes specified in this study. i think a lot of them are pertinent to the subject i am only worried as to the amount of work involved. Also, is it possible that some outcomes in particular might be affected by the healthcare system? for example, neonatal ICU. is it possible that certain facilities do not have access to or differing access to neonatal ICU. or neonatal morbidity where duration of neonatal hospital stay might differ. These factors might make the result less easy to interpret. maybe some outcomes can be excluded to make the work less arduous. This is only a suggestion and i would feel totally fine if the authors decided to include all of the mentioned outcomes. On a side note, when i look at this objective, what comes into my mind in term of study design is more of an epidemiological study, ecological study. Are some of these outcomes part of a systematic data collection and reporting system throughout Africa that can be leveraged in some sort. again just a suggestion. 2. exclusion criteria: have the authors considered maybe there are some multicentered study where African patients were included. will they try and contact the authors to get access to that part of the data?

	Risk of Bias: You will include RCTs. which risk of bias tool will you use for RCTs. also RCTs tend to skew the outcome due to the intervention. Do you plan on only including the outcomes of the control group? Analysis: will you by stratifying and meta-analysing based on the type of HIP?

REVIEWER	giorgina b piccolo CH Le Mans France University of Torino Italy
REVIEW RETURNED	03-Jun-2020

GENERAL COMMENTS	thank you for letting me evaluate this ambitious protocol. I have some points to be addressed: 1. Why Africa? please explain better why this is crucial (no study is not enough) 2. the limit to Africa has to be highlighted in the objectives and throughout the paper 3. please review and discuss the definitions of preeclampsia 4. let me suggest to include Embase in the databases. the definition of databases has to be improved 5. please use the MOOSE more than PRISMA: these are likely to be observational studies

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
1. Outcomes.
There are a lot of outcomes specified in this study. i think a lot of them are pertinent to the subject i am only worried as to the amount of work involved.
Also, is it possible that some outcomes in particular might be affected by the healthcare system? for example, neonatal ICU. is it possible that certain facilities do not have access to or differing access to neonatal ICU. or neonatal morbidity where duration of neonatal hospital stay might differ. These factors might make the result less easy to interpret. maybe some outcomes can be excluded to make the work less arduous. This is only a suggestion and i would feel totally fine if the authors decided to include all of the mentioned outcomes.

Thank you for the comment. We agree that the multiple outcomes are a challenge when considering data collection and analysis. We have considered increasing the review team. We will also discuss the limitation imposed by health system when it comes to interpreting these results as suggested by the reviewer.

On a side note, when i look at this objective, what comes into my mind in term of study design is more of an epidemiological study, ecological study. Are some of these outcomes part of a systematic data collection and reporting system throughout Africa that can be leveraged in some sort. again just a suggestion.

We thank you for the comment. We are not going to use routinely collected data from government bodies as this may be time consuming, but we appreciate how much this may add to the study.

2. exclusion criteria:
have the authors considered maybe there are some multicentered study where African patients were included. will they try and contact the authors to get access to that part of the data?

Thank you for the suggestion. Yes, if we come across any multicentre studies, we will contact authors so we may get the data on Africa. We are aware of one multi center study which investigated obesity in children born from GDM pregnancies, but it only included less than 10 children from Kenya and South Africa.

Risk of Bias:
You will include RCTs. which risk of bias tool will you use for RCTs. also RCTs tend to skew the outcome due to the intervention. Do you plan on only including the outcomes of the control group?

Thank you for the comment. We will use the Cochrane Risk of bias tool for RCTs. Yes, we plan to include outcomes from control groups, especially if they are from cohort studies.

Analysis:
will you by stratifying and meta-analysing based on the type of HIP?

Thank you. Yes, we will do stratum-specific estimates where data are sufficient.

Reviewer: 2
giorgina b piccoli
CH Le Mans France
University of Torino Italy

Please state any competing interests or state 'None declared':
none declared

Comments to the Author
thank you for letting me evaluate this ambitious protocol.
I have some points to be addressed:
1. Why Africa? please explain better why this is crucial (no study is not enough)
Thank you for the comment. Africa is a continent where many countries are experiencing an increasing prevalence of non-communicable diseases (NCDs), while burdens from infectious diseases continue to be high and health systems remain weak. This makes it imperative to consider the outcomes of these pregnancies in Africa alone.

2. the limit to Africa has to be highlighted in the objectives and throughout the paper

Thank, we have done so.

3. please review and discuss the definitions of preeclampsia

Thank you, we have revised this to 1. Preeclampsia defined as a multisystem progressive disorder characterised by the new onset of hypertension and proteinuria or the new onset of hypertension and significant end-organ dysfunction with or without proteinuria in the last half of pregnancy or postpartum.24 The end-evidence of organ dysfunction include: i. proteinuria ≥0.3 g in a 24-hour urine specimen or protein/creatinine ratio ≥0.3 (mg/mg) (30 mg/mmol) in a random urine specimen or dipstick ≥2+ if a quantitative measurement is unavailable, ii. renal insufficiency- serum creatinine >1.1 mg/dL (97.2 µmol/L) or doubling of the creatinine concentration in the absence of other renal disease, iii. liver involvement- liver transaminases at least twice the upper limit of the normal concentrations for

the local laboratory, iv. neurological complications- new-onset and persistent headache not accounted by alternative diagnoses and not responding to usual doses of analgesics, and visual symptoms (blurred vision, flashing lights or sparks, scotomata), and v. thrombocytopaenia- platelet count <100,000/µL.24, 25

4. let me suggest to include Embase in the databases. the definition of databases has to be improved

Thank you, we have done so.
5. please use the MOOSE more than PRISMA: these are likely to be observational studies

Thank, we have made the change in the protocol

VERSION 2 – REVIEW

REVIEWER	Han Ting Wang Maisonneuve-Rosemont hospital, Montreal, Quebec, Canada
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	I believe the authors have adressed well all the comments and concerns formulated by reviewers. After this reading, i do believe this manuscript should be accepted for publication. Good luck in this ambitious work!

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Han Ting Wang
Institution and Country: Maisonneuve-Rosemont hospital,
Montreal, Quebec, Canada
Please state any competing interests or state ‘None declared’: None declared

Comments to the Author
I believe the authors have adressed well all the comments and concerns formulated by reviewers.
After this reading, i do believe this manuscript should be accepted for publication.
Good luck in this ambitious work!

Author’s Response: Thank you.